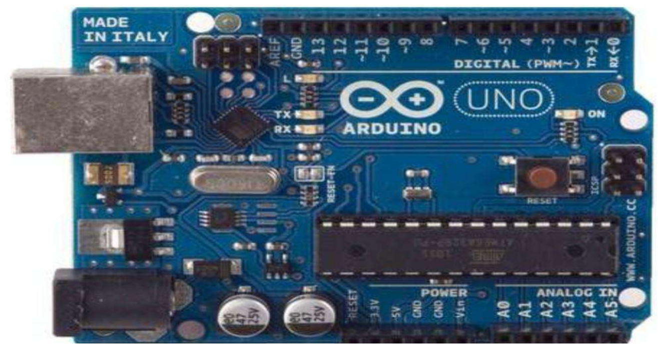
TITLE: Automatic speed control and accident avoidance system using multi sensors

Abstract

- The increasing number of road accidents due to driverfatigue, obstacles, and hazardous conditions necessitates the development of advanced safety systems in vehicles.
- This project proposes an integrated system that employs ultrasonic sensors, eye blink sensors, and smoke sensors to enhance vehicular safety through automatic speed control and accident avoidance mechanisms.

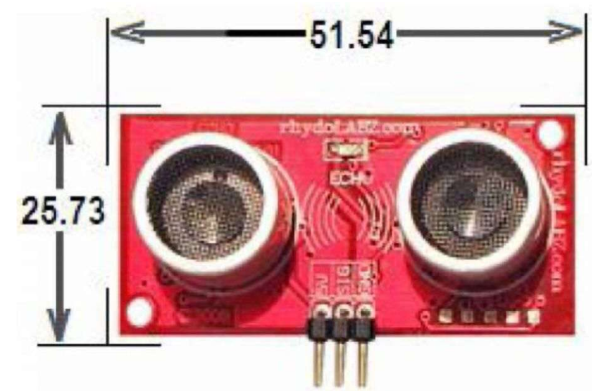
Components Utilized



ARDUINO UNO CONTROLLER:

The Arduino Uno is a microcontroller board based on the ATmega328P chip. It has 14 digital I/O pins, including 6 with PWM (Pulse Width Modulation) output, and 6 analog input pins (A0-A5).

The operating voltage is 5V, and the recommended input voltage is 7-12V, with a limit of 6-20V.



ULTRASONIC SENSOR:

This "ECHO" Ultrasonic Distance Sensor from Rhydolabz is an amazing product that provides very short (2CM) to long-range (4M) detection and ranging. The sensor provides precise, stable non- contact distance measurements from 2cm to 4 meters with very high accuracy. Its compact size, higher range and easy usability make it a handy sensor for distance measurement and mapping.

DC motor:

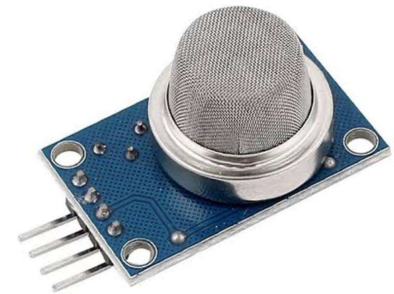
DC motors are configured in many types and sizes, including brush less, servo, and gear motor types. A motor consists of a rotor and a permanent magnetic field stator. The magnetic field is maintained using either permanent magnets or electromagnetic windings. DC motors are most commonly used in variable speed and torque.

LCD:

Liquid Crystal Display also called as LCD is very helpful in providing user interface as well as for debugging purpose. The most common type of LCD controller is HITACHI 44780 which provides a simple interface between the controller & an LCD. These LCD's are very simple to interface with the controller as well as are cost effective.

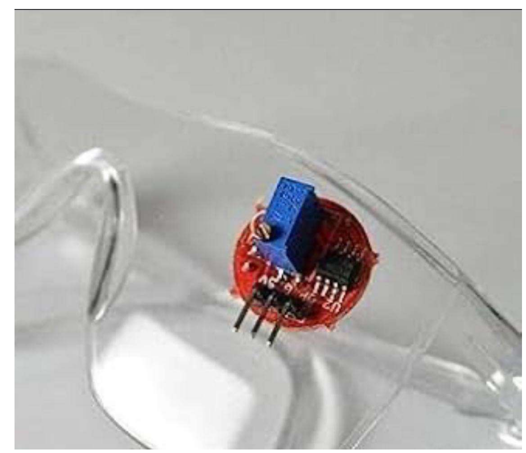
SMOKE SENSOR:

They are used in gas leakage detecting equipments in family and industry, are suitable for detecting of LPG, natural gas , town gas, avoid the noise of alcohol and cooking fumes and cigarette smoke.

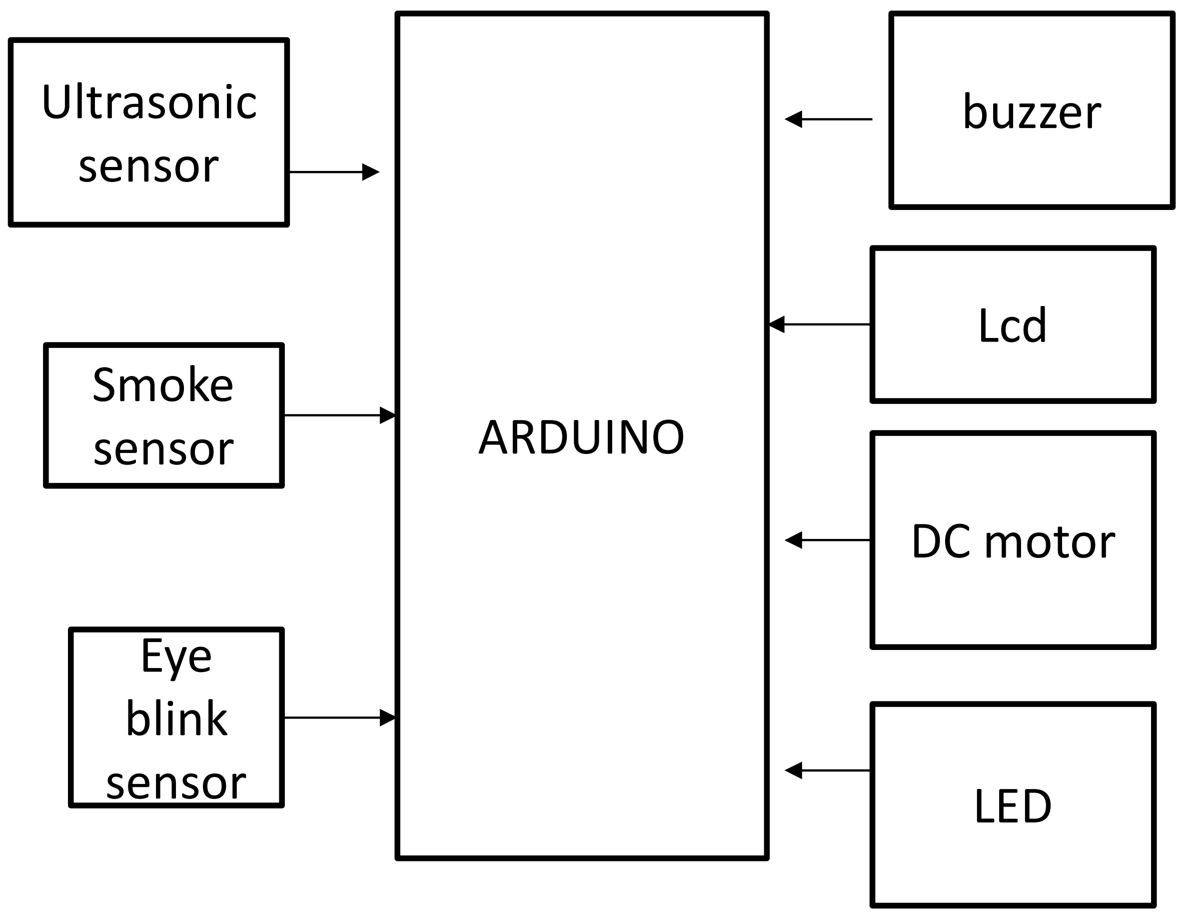


EYE BLINK SENSOR

The eye-blink sensor works by illuminating the eye and eyelid area with infrared light, then monitoring the changes in the reflected light using a phototransistor and differentiator circuit. The exact functionality depends greatly on the positioning and aiming of the emitter and detector with respect to the eye.



Block Diagram



Code

EYE BLINK

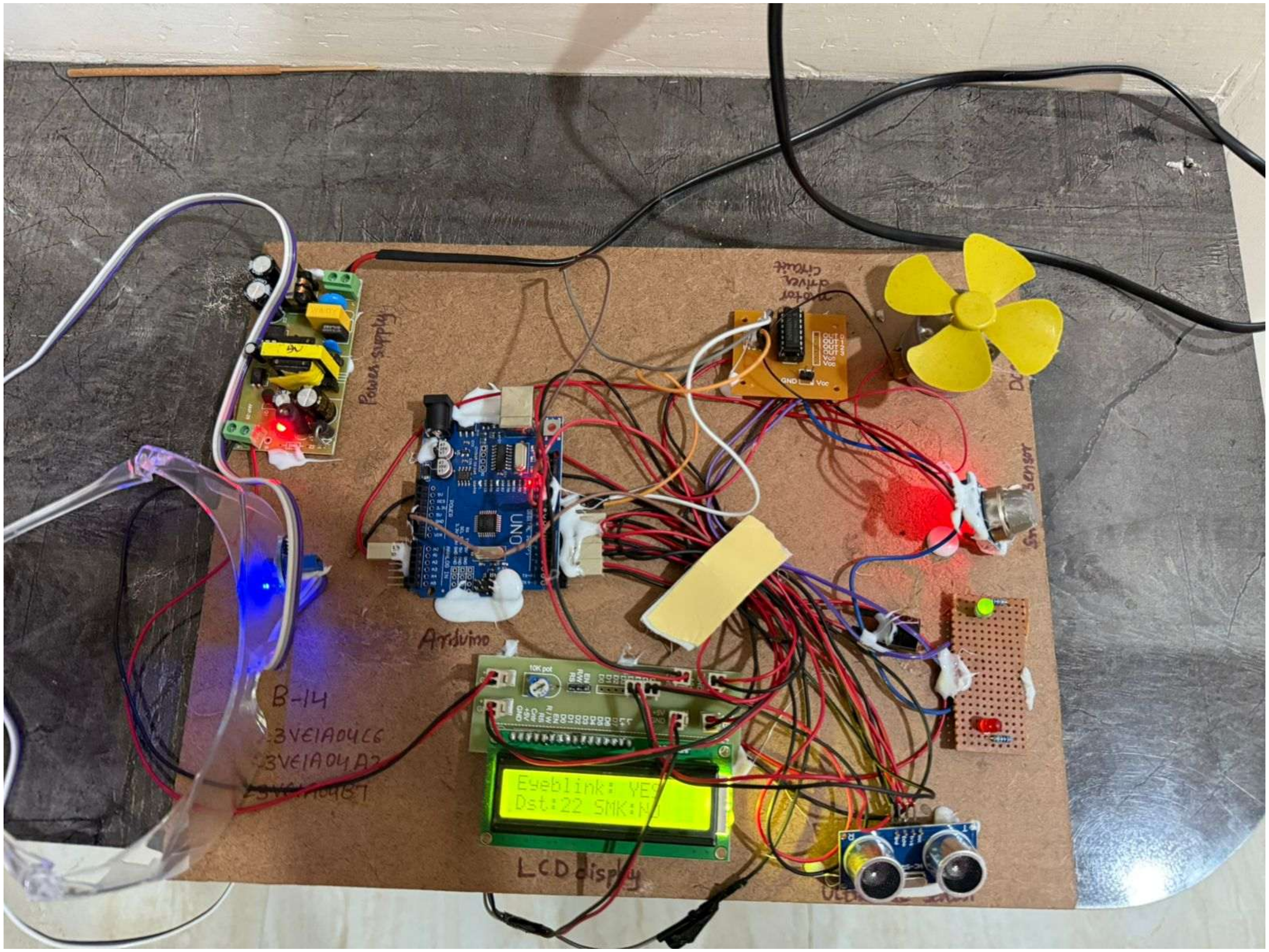
```
// Pin configuration
const int smokeSensorPin = A0;      // MQ-3 Smoke sensor (Analog)
const int eyeBlinkSensorPin = 2;      // Eye Blink sensor (Digital)
const int relayPin = 8;               // Relay to control motor
const int buzzerPin = 9;              // Buzzer pin
const int smokeThreshold = 300;     // Adjust after testing


unsigned long lastBlinkTime = 0;
const unsigned long blinkTimeout = 10000; // 10 seconds without blink
```

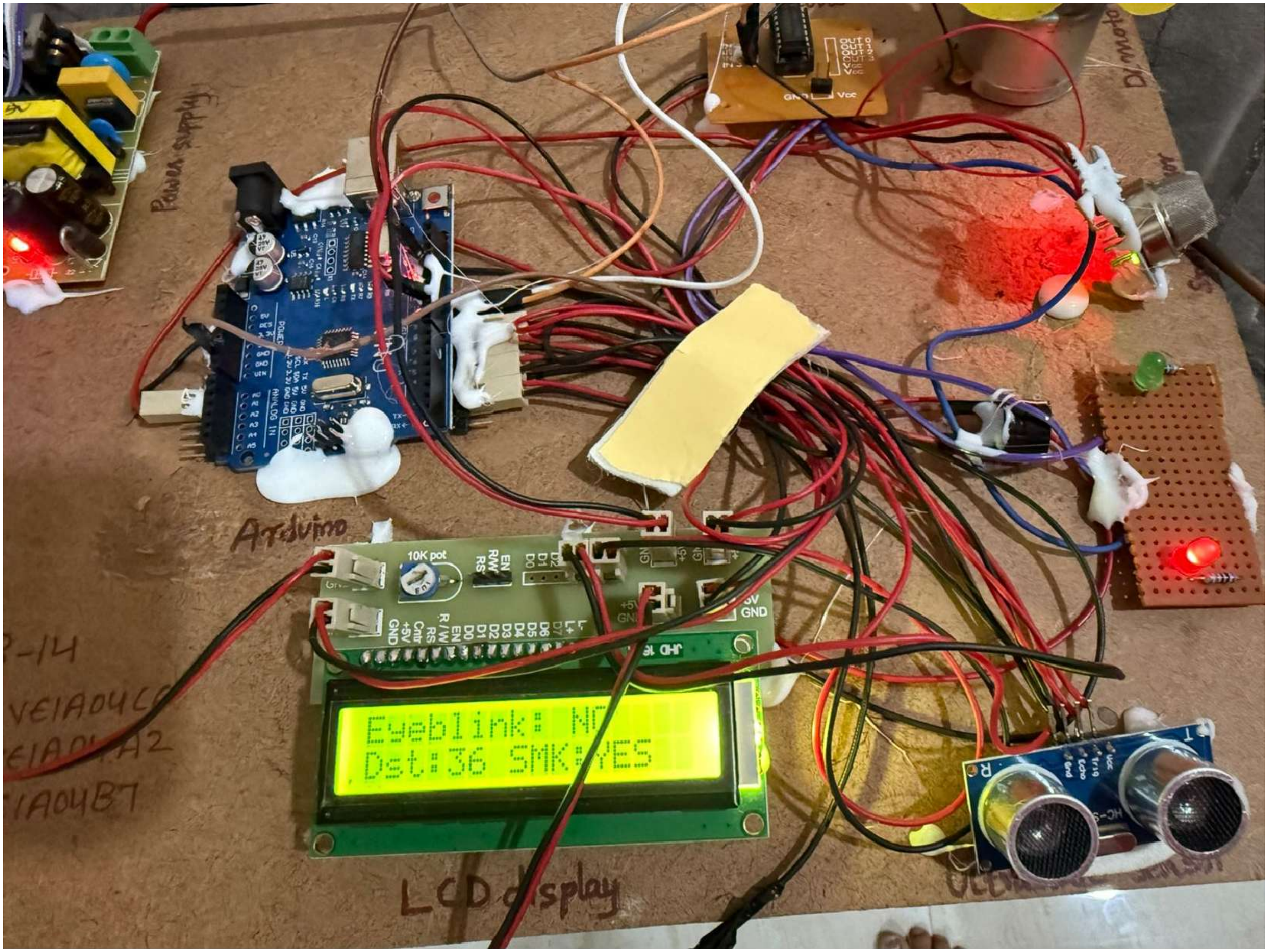
```
void setup() {
  pinMode(smokeSensorPin, INPUT);
  pinMode(eyeBlinkSensorPin, INPUT);
  pinMode(relayPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  digitalWrite(relayPin, HIGH);  // Motor ON
  digitalWrite(buzzerPin, LOW);  // Buzzer OFF

  Serial.begin(9600);
  Serial.println("Vehicle Safety System Initialized");
}

void loop() {
  int smokeValue = analogRead(smokeSensorPin);
  int eyeBlinkStatus = digitalRead(eyeBlinkSensorPin);

  Serial.print(“smoke: ");
  Serial.print(smokeValue);
  Serial.print(" | Eye Blink: ");
  Serial.println(eyeBlinkStatus);
```

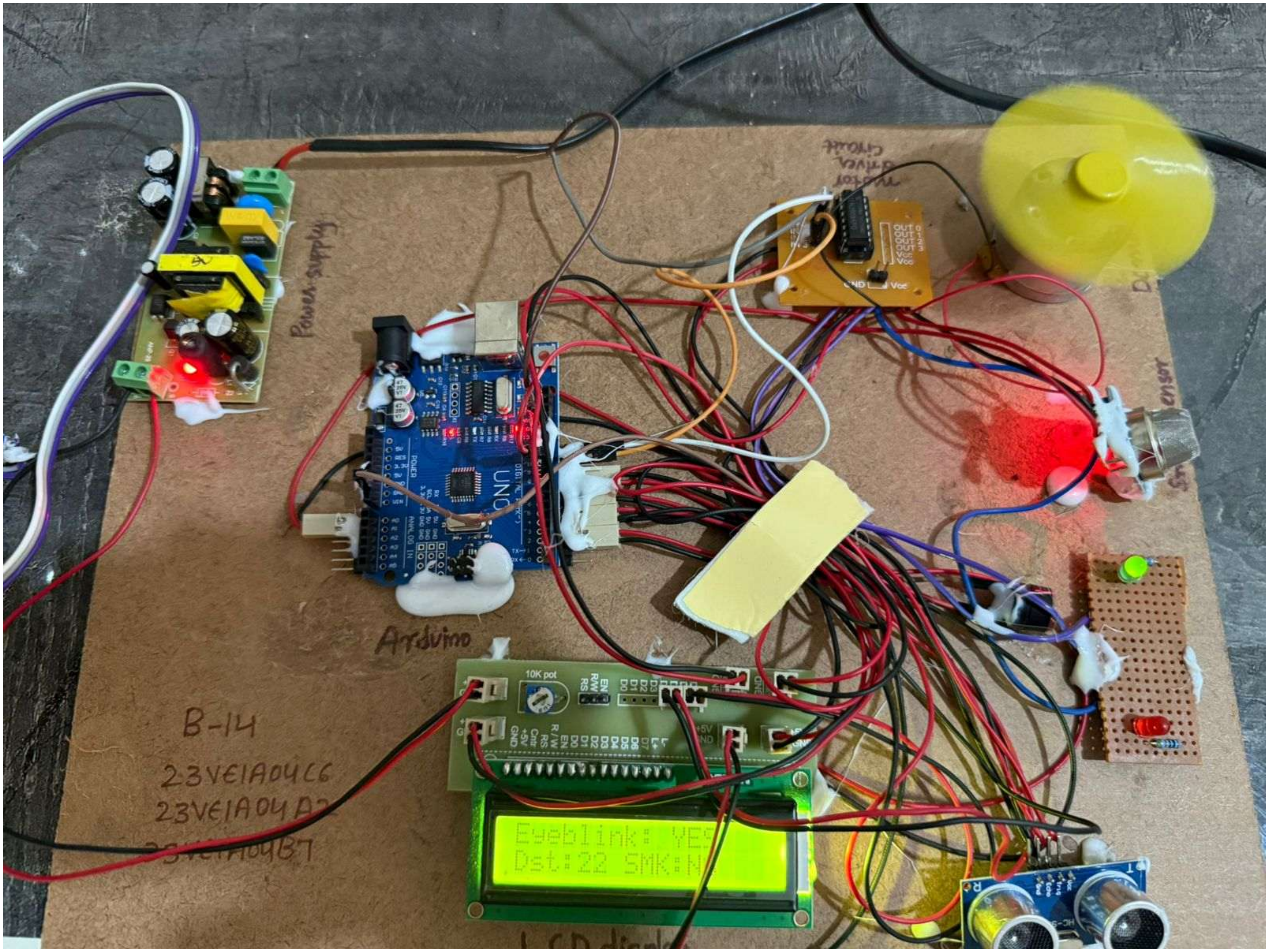
```
// Case 1: Smoke Detected
 if (smokeValue > smokeThreshold) {
  Serial.println("SMOKE DETECTED! Stopping vehicle...");
  digitalWrite(relayPin, LOW);   // Motor OFF
  digitalWrite(buzzerPin, HIGH); // Buzzer ON
 }

 // Case 2: Eye blink normal
 else if (eyeBlinkStatus == LOW) {
  lastBlinkTime = millis();     // Blink detected
  digitalWrite(relayPin, HIGH); // Motor ON
  digitalWrite(buzzerPin, LOW); // Buzzer OFF
 }

 // Case 3: Drowsiness detected
 else {
  if (millis() - lastBlinkTime > blinkTimeout) {
   Serial.println("DROWSINESS DETECTED! Stopping vehicle...");
```

```
digitalWrite(relayPin, LOW);   // Motor OFF
      digitalWrite(buzzerPin, HIGH); // Buzzer ON
    } else {
      digitalWrite(relayPin, HIGH);  // Motor ON
      digitalWrite(buzzerPin, LOW);  // Buzzer OFF
    }
  }

  delay(500);
}
```

Simulation Results




Power supply
Arduino
10K pot
LCD display
Eyeblink: N
Dst:36 SMK:YES



Power supply
Arduino
UNO
10K pot
B-14
23VE1A04C6

Advantages

- **Enhanced Road Safety**
  - Automatically detects obstacles, driver fatigue, and fire hazards—significantly reducing accident risks.
- **Low Cost and Easy Implementation**
  - Uses inexpensive and widely available sensors and microcontrollers like Arduino, making it suitable for educational and prototype use.
- **Low Power Consumption**
  - Efficient design ensures minimal energy use, supporting use in battery-powered embedded systems.
- **Compact and Lightweight**
  - The design occupies little space, making it ideal for integration into existing vehicles without major modifications.

Limitations

- **Limited Detection Range**

Ultrasonic sensors have a relatively short range (typically < 4 meters), which may not be sufficient at high speeds.

- **False Alarms from Eye Blink Sensor**

Rapid blinking or head movements may cause incorrect detection of drowsiness, leading to unnecessary stops or alerts.

- **Environmental Interference**

Rain, dust, smoke, or reflective surfaces may affect the accuracy of the ultrasonic and smoke sensors.

- **Limited Fire Detection Capability**

The smoke sensor may not detect all types of fire or chemical hazards accurately, especially in well-ventilated or open areas.

Applications

- **Passenger Vehicles**

Enhances safety in cars by monitoring driver drowsiness, obstacles, and fire hazards.

- **Commercial Trucks and Buses**

Helps prevent fatigue-related accidents on long routes by alerting drivers and slowing/stopping the vehicle.

- **School and College Transport Vehicles**

Improves safety for student transport by monitoring driver alertness and vehicle surroundings.

- **Emergency Vehicles**

Can be used in ambulances or fire trucks to ensure safety while traveling at high speeds

Future scope

- **Voice Alerts and Multilingual Support**

Add voice-based warnings in different languages for better accessibility and driver comprehension.

- **Automatic Braking and Steering Control**

Connect with advanced braking systems and electronic steering for full autonomous emergency handling

- **Vehicle-to-Vehicle (V2V) Communication**

Enable communication between vehicles to share hazard or traffic data and further reduce collision risks

Conclusion

The proposed system effectively demonstrates how embedded technology and sensors can be combined to enhance vehicle safety through automatic speed control and accident avoidance. By integrating an ultrasonic sensor for obstacle detection, an eye blink sensor for monitoring driver alertness, and a smoke sensor for fire hazard detection, the system offers a proactive solution to reduce the likelihood of accidents.

While the system has limitations—such as environmental interference and range constraints—it serves as a strong foundation for future advancements in intelligent transportation systems. With further development and integration with real-time communication modules and vehicle control systems, this prototype can be evolved into a comprehensive safety solution for modern vehicles.

Ultimately, this project highlights the potential of low-cost, real-time safety mechanisms in saving lives and making road travel safer

Thank you